Govt. Graduate College of Science, Wahdat Road, Lahore
Faculty of Computing & Information Technology (FCIT), University of the Punjab, Lahore

FINAL YEAR PROJECT (FYP)

PROPOSAL DOCUMENT

"Smart Academic Communication Platform (SACP)"

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FACULTY OF COMPUTING & INFORMATION TECHNOLOGY (FCIT), UNIVERSITY OF THE PUNJAB, LAHORE

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1 Introduction

1.1 Project Title

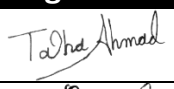
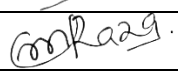
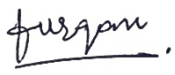
Smart Academic Communication Platform (SACP)

1.2 Project Overview Statement

Project Title: Smart Academic Communication Platform (SACP)

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Project Members:

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Project Goal:

Academic institutions often rely on inefficient ways of communication to interact with faculty members and students. These methods are generally done manually, and they also require a considerable amount of time, are subject to human error, and make it difficult for academic institutions to effectively send important and timely messages such as attendance warnings, exam schedules, and other important notifications to thousands of students and many faculty members within a reasonable timeframe. In this manual process, many students won't receive updates early and some might miss important announcements.

The **Smart Academic Communication Platform (SACP)** implements an automated solution that is a data analysis-driven, intelligent solution to address several operational obstacles that the academic community faces today. It works directly with academic records to allow faculty to activate personalized, contextually referenced workflow to trigger a specific action in multiple ways, as well as provide detailed analytics in order to increase the overall efficiency and decision making. It also provides rule-based automation and machine learning techniques for personalized and context-aware communication. It introduces a system where all possible manual work is automated and customizable.

Objectives

Sr. No.	Objectives
1.	Develop a centralized communication platform for academic institutions.
2.	Implement dynamic message generation via templates for personalization.
3.	Provide automated, rule-based communication workflows triggered by academic events (e.g., attendance shortage, assignment deadlines, announcements, etc.)
4.	Enable bulk communication through campaigns and scheduled announcements.
5.	Support targeted communication based on academic records.
6.	Develop an analytics dashboard to monitor communication performance.
7.	Maintain communication logs and message delivery tracking.
8.	Implement machine learning-based email classification to prioritize important messages and filter-out spam messages.
9.	Provide 1-to-1 real-time communication (chat system) within the platform.

Project Success Criteria:

The success criteria of this project can be determined by the following factors, if:

- The system can automate academic communication based on data and user-defined rules.
- Admins can send personalized messages to large groups both manually and via rule-based automation.
- The platform stores communication logs and delivery tracking.
- Machine learning properly filter-out messages and prioritize them.
- Provides smooth communication workflow compared to traditional methods.
- The system provides usability and scalability in an academic institution.

Assumptions, Risks, and Obstacles:

- Academic data integration may require structured data formats.
- Message delivery services may experience delays depending on server load.
- High communication traffic could affect the performance if not optimized.
- Making template and rule-based automation engine might be complex and difficult to implement without proper architecture and planning.
- Access control and data privacy must be carefully implemented.
- For effective training of the machine learning classification model, large datasets are required.

Type of Project:

- ✓ Development

Target End Users:

- Academic Administrators
- Faculty Members
- University Students

Development Technology:

- ✓ Object-Oriented Development

Platform:

- ✓ Web-based
- ✓ Distributed System Architecture

Suggested Project Supervisor:

Approved By:

Date:

1.3 Project Goals and Objectives

Goals:

The main goal of the Smart Academic Communication Platform (SACP) is to create an intelligent communication platform for academic institutions that simplifies large-scale academic messaging while ensures timely delivery of important, relevant, and personalized communication across the institution.

Objectives:

- To make academic communication easy and efficient.
- To develop a template-based communication system.
- To enable personalized messages to large number of students and faculty members.
- To implement automated rule-based messaging workflow.
- To provide campaign management features for efficient institutional announcements.
- To maintain detailed logs of all communication activities.
- To implement machine learning based services for emails classification and prioritization.

- To provide analytics for monitoring and improving communication workflows.
- To enhance communication efficiency by reducing manual efforts.

1.4 High-Level System Components

The Smart Academic Communication Platform will consist of following high-level (core and optional) modules:

1. **Authentication and Authorization**
   1.1. User login and secure authentication.
   1.2. Role based access control via authorization.
2. **Academic Data Management**
   2.1. Stores academic records to make workflows better and effective.
   2.2. Course and department information for selecting the right audience.
3. **Template Management System**
   3.1. Creation of different templates for sending repeated and large number of messages.
   3.2. Provides dynamic placeholders for personalized communication.
4. **Campaign Management**
   4.1. Provides campaign creation and scheduling workflows.
   4.2. Selecting the target audience easily based on records in the database.
5. **Automation Rule Engine**
   5.1. Event-based based on defined rules communication triggers.
   5.2. Provides user-friendly interface for creating automated workflows for student risk management, daily announcements, etc.
6. **Messaging Service and Notification System**
   6.1. Provides email delivery and tracking system.
   6.2. Provides campaign alerts and remainders of important events.
7. **Real-Time Communication Module**
   7.1. Provides one-to-one and optionally group chat functionality.
   7.2. Making the feedback communication centralized inside the platform.
8. **Email Classification and Prioritization System**
   8.1. Provides a system which uses machine learning classifier model to filter out important and spam emails.
   8.2. Prioritizing of critical academic messages for better attention and response.
9. **Analytics and Reporting**
   9.1. Provides a dashboard for administrators and faculty members to track and visualize communication metrics and statistics of campaigns.
10. **File and Document Sharing Module**
    10.1. Provides support for attachments in messages.

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10.2. Enable document sharing among users.

1.5 List of Optional Functional Units

- Add support for multi-channel communication like WhatsApp and SMS messaging.
- AI-assisted message/template generation and summarizer for long messages.
- AI-based campaign recommendations and assistance.

1.6 Exclusions

The following features are outside the scope of this project:

- Financial or payment systems.
- Learning Management System (LMS) functionality.
- A chat application for public like WhatsApp or Messenger.

1.7 Application Architecture

The system will follow a 3-tier architecture.

Presentation Layer (Frontend):

It'll be responsible for providing a platform for user interaction by using following technologies:

- React
- HTML, CSS, JavaScript

Application Layer (Backend & AI):

Its backend part will be responsible for core business logic, messaging services, data management, etc. There're two main parts:

- **Backend:** It's responsible for all core and real-time services including system heavy processing. It'll be using .NET (C#) framework.
- **AI Layer:** This part is responsible for all kinds of AI/ML services. It'll be using Python.

Database Layer:

It'll be responsible for storing data of all entities and managing relationships between them by using following technologies:

- PostgreSQL / Azure SQL

1.8 Hardware and Software Specifications

Hardware:

- Intel Core i5 or higher
- 8-16 GB RAM
- 80-128 GB SSD
- Internet Connectivity

Software:

- Windows (10/11) / Linux / macOS
- Node.js
- .NET SDK
- Python

1.9 Tools and Technologies Used

Development Tools:

The following IDEs provide strong support for specific development goals:

- Visual Studio Code / Visual Studio
- Pycharm

The following tools provide user-friendly platform for API testing:

- Swagger / Postman

The following tools provide strong support for version control to handle different conflicts while building the project:

- Git / GitHub

The following tools will be used for project management, designing and documentation writing:

- Jira / Trello / MS Project
- MS Office
- Draw.io, PlantUML, ERD Plus
- Figma / Adobe XD

Technologies:

The technology will be used for frontend development, which provides responsive and interactive user interfaces, is:

- React
- TailwindCSS / Bootstrap / Material UI
- Redux
- Charts.js / Recharts

The technologies will be used for creating scalable and large backend system and ML models, are:

- Backend
    - C#
    - ASP.NET Core Web APIs
    - Entity Framework + LINQ
    - JWT Auth
    - Hangfire / Quartz.NET
    - SignalR
    - Redis
- AI Layer
    - Python
    - FastAPI / Flask
    - Scikit-Learn
    - Pandas
    - NumPy
    - spaCy / NLTK
    - Matplotlib
    - Seaborn
    - Jupyter Notebook

The database management system that will be used in this project is:

- PostgreSQL / Azure SQL